

Boston Food Safety Inspections Trend (2005-2026): Volume, Severity, and Neighborhood Change

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Background

Boston's health inspections spans 20+ years of restaurant data. The public data includes outcomes on inspections, pass or fails and violation levels, which can be good to reveal public health trends and neighborhood disparities.

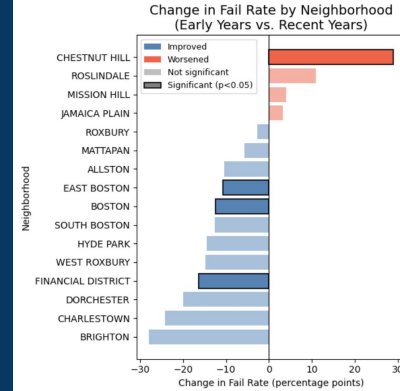
Research Question

To what extent have Boston **food safety inspection outcomes** changed over **time** in terms of **volume** and **severity**, and are certain **neighborhoods** improving or worsening over the past 20 years?

Data & Method

- **Source:** Open public data portal of Boston Inspectional Services Department food inspection records, 2005-2026.
- **Unit of observation:** Raw data has one row per violation. Rows were grouped by business name and inspection date to reconstruct one row per inspection, with a new column counting violations per inspection.
- **Outcome variable:** Result column standardized and merged to only contain "Pass" and "Fail" All other outcomes dropped.
- **Severity:** Violation level (* / ** / ***) converted to numeric (1 / 2 / 3) to enable quantitative analysis. Missing severity (rows with no violation, i.e. passing inspections) excluded from severity calculations but kept in fail rate calculations.
- **Neighborhood:** City/neighborhood labels cleaned and unified. Neighborhoods with fewer than 200 inspections total excluded as estimates were too noisy.
- **Time:** Inspection date parsed to extract year. Rows with unparseable dates dropped. Analysis spans 2005-2026.

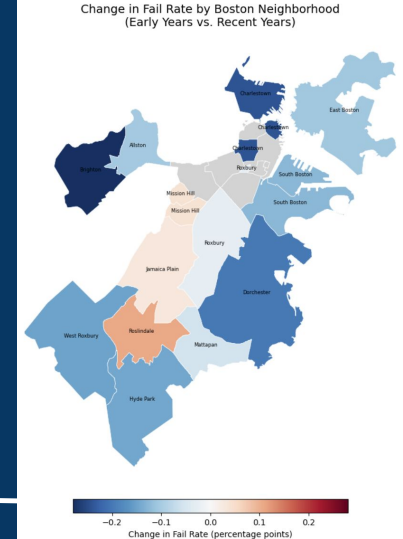
Neighborhood change scores



Neighborhood level regression -> 3 neighborhoods with statistically significant declining trends (Brighton, Charlestown, Dorchester) + 1 with significant worsening (Chestnut Hill).

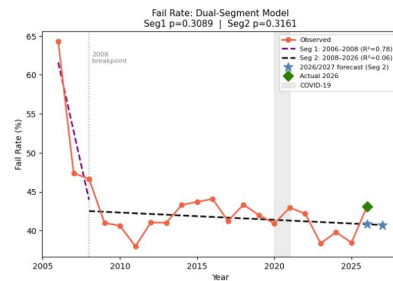
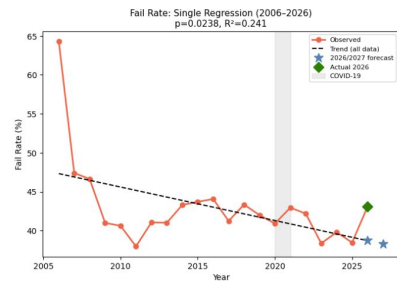
Most Boston neighborhoods saw a decline in fail rates over the past 20 years, with Brighton, Charlestown, and Dorchester showing the largest improvements. Gray neighborhoods had insufficient inspection data for reliable estimates

Choropleth map (change in fail rate)

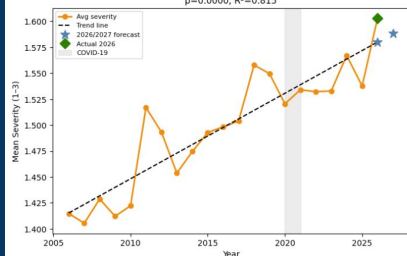


Citywide trends over time – w/ 2026 Prediction & 2027 Forecast

Dual-Segment Fail Rate Model vs Single Regression



Avg Severity ~ Year
p=0.0000, R²=0.815



Model Comparison: Fail Rate & Severity Predictions

	Fail: Single	Fail: Seg 2	Severity
Time Coef. (pp/yr)	-0.4303	-0.0941	0.0082
R²	0.241	0.059	0.815
Pred 2026	36.71%	40.82%	1.5801
Pred 2027	38.28%	40.73%	1.5884
Actual 2026	43.07%	43.07%	1.6030
Error	-4.36 pp	-2.25 pp	-0.0228

Model Fit

- **Severity model fits well** ($R^2 = 0.815$), = consistent with linear trend over time
- **Fail Rate Single Regression fits more poorly** ($R^2 = 0.241$), likely driven by early-year outliers (2005-2007)
- **Fail Rate Dual-Segment model is better suited for our goals**
-> the 2nd segment captures the plateau era used for predictions

Why the Dual-Segment (Seg 2) Model is Preferred for Forecasting

- **Smaller prediction error:**
-> Seg 2 model underestimates 2026 by 2.25 pp = half the error of the single regression model (4.36 pp)
- **More realistic slope:** Visually, the Seg 2 trend better reflects recent data (by excluding decade-old outliers that no longer represent current dynamics)

- **Low R^2 is actually expected:** The weaker R^2 in Seg 2 (0.059) is not a bad sign necessarily, it's an indicator that year alone does not strongly predict fail rate variation once the data plateaus

Limitations

- Small neighborhoods -> noisy estimates
- COVID-19 distorts 2020-2021 as we see inspection volumes drop
- Confounders to consider: enforcement bias, language barriers, inspector behavior
- Independence assumption for neighborhoods might not be correct as our models does not include not include neighborhoods

Conclusion

- **Boston's citywide fail rate declined over 20 years** ($p = 0.024$), though single regression fit is not ideal ($R^2 = 0.241$) due to early outliers (2005-2007) => **dual-segment model** captures the post-2008 plateau and cuts prediction error by almost half (2.25 pp vs. 4.36 pp)
- **Average violation severity increased significantly** over the same period ($R^2 = 0.815$, $p < 0.001$) => more serious violations
- **Neighborhood level = 3** improved significantly and 1 worsened
- **Both models underestimated** the 2026 fail rate & severity => likely post-COVID rebound in violation activity not captured by linear trend
- **Limitations:** unequal neighborhood sample sizes + potential enforcement and reporting biases across years and districts

Reference

City of Boston Inspectional Services Department. (2026). Food establishment inspections [Dataset]. Analyze Boston / Department of Innovation and Technology. <<https://data.boston.gov/dataset/food-establishment-inspections>> extracted on February 15 2026

Notes: Severity model only on the subset of data of failed inspections (passed inspections do not have severity levels)
+ Both models are simple linear regressions with year as the only predictor